Probability rules

1. Compliment rule:

$\Rightarrow P(A) = 1 - P(A')$ or $P(A') = 1 - P(A)$

2. Mutually exclusive rule:

$\Rightarrow$ If $P(A \cap B) = 0$, then events A and B are mutually exclusive.

$\Rightarrow$ If $P(A \cap B) \neq 0$, then events A and B are not mutually exclusive.

3. Collectively eschaustive events:

→ Events A_1, A_2, A_3 are collectively eschaustive when one of these events must occur when we select from the sample space $\{A_1, A_2, A_3\} = S$

→ Collectively they take up the sample sample.

4. Addition rule:

$\Rightarrow P(A \cup B) = P(A) + P(B) + P(A \cap B)$

If A and B are mutually exclusive then

$\Rightarrow P(A \cup B) = P(A) + P(B)$

(5) Conditional probability rule:

$$\Rightarrow P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{or} \quad P(B|A) = \frac{P(A \cap B)}{P(A)}$$

(6) Multiplication rule:

$$\Rightarrow P(A \cap B) = P(A|B)P(B) \quad \text{or} \quad P(A \cap B) = P(B|A)P(A)$$

(7) Independence rule:

$$\Rightarrow P(A \cap B) = P(A)P(B)$$

$$\Rightarrow P(A|B) = P(A) \quad \text{and} \quad P(B|A) = P(B)$$

If these conditions apply, the events A and B are statistically independant

(8) Bayes' rule:

$$\Rightarrow P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A')P(A')} = \frac{P(B|A)P(A)}{P(B)}$$

or

$$\Rightarrow P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B')P(B')} = \frac{P(A|B)P(B)}{P(A)}$$

9 Total probability rule:

$$P(A) = P(A|B)P(B) + P(A|B')P(B')$$

or

$$P(B) = P(B|A)P(A) + P(B|A')P(A')$$

Probability "tools"

1 Contingency table

	B	B'	Total
A	$P(A\cap B)$ $= P(A\mid B)P(B)$ $= P(B\mid A)P(A)$	$P(A\cap B')$ $= P(A\mid B')P(B')$ $= P(B'\mid A)P(A)$	$P(A)$
A'	$P(A'\cap B)$ $= P(A'\mid B)P(B)$ $= P(B\mid A')P(A')$	$P(A'\cap B')$ $= P(A'\mid B')P(B')$ $= P(B'\mid A')P(A')$	$P(A')$
Total	$P(B)$	$P(B')$	1

Examples of rules:

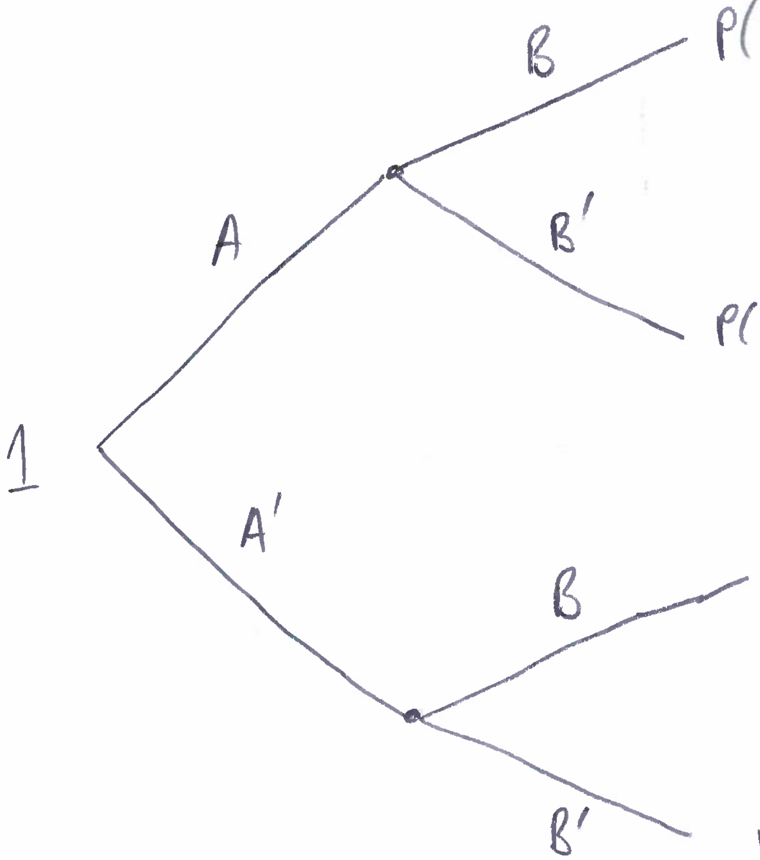
(a) $P(A) = 1 - P(A')$

(b) $P(A\cap B) = P(A\mid B)P(B) = P(B\mid A)P(A)$

(c) $P(A) + P(A') = 1$

(d) $P(A) = P(A\mid B)P(B) + P(A\mid B')P(B')$

2 Tree diagram



$$P(A \cap B) = P(A|B)\,P(B) = P(B|A)\,P(A)$$

$$P(A \cap B') = P(A|B')\,P(B') = P(B'|A)\,P(A)$$

$$P(A' \cap B) = P(A'|B)\,P(B) = P(B|A')\,P(A')$$

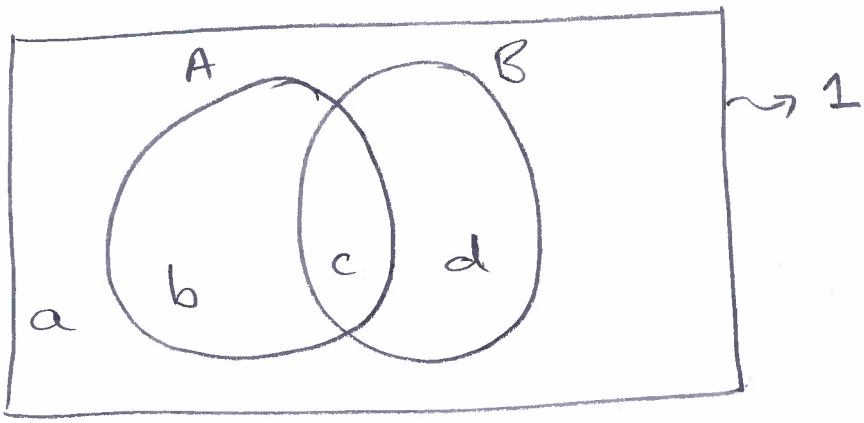
$$P(A' \cap B') = P(A'|B')\,P(B') = P(B'|A')\,P(A')$$

Examples of rules:

(a) $P(A) = P(A \cap B) + P(A \cap B')$

(b) $P(B) = P(A \cap B) + P(A' \cap B)$

3 Venn diagram



Note events:

$\{b, c\} \Rightarrow A$

$\{c, d\} \Rightarrow B$

$\{a, d\} \Rightarrow A'$

$\{a, b\} \Rightarrow B'$

$\{c\} \Rightarrow A \cap B$

$\{a\} \Rightarrow A' \cap B'$

$\{b\} \Rightarrow A \cap B'$

$\{d\} \Rightarrow A' \cap B$